Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

Residence Information
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UNITED STATES

Mailing Address
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Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**Thermal System Efficiency Optimization via Spectral Analysis of Heat-Load Frequencies and Phase-Locked Cooling Actuation**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**
Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:
Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 1

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

1. **HeadySystems Inc.**
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Applicant	Organization
Applicant type	Assignee

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

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Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	01/22/2026

PROVISIONAL PATENT APPLICATION

System Efficiency and Thermal Optimization via Spectral Phase Alignment (HeadyCool)

(Specification and Drawings)

Applicant / Assignee	HeadySystems Inc.
Inventor	Eric Haywood
Correspondence Address	149 Remington Street Unit 425, Fort Collins, CO 80524
Entity Status	Small Entity
Government Interest	None
Attorney Docket	PPA-029
Date	January 22, 2026

Title

PPA-029: HeadyCool — Spectral Thermal-Cycle Optimization via Phase-Aligned Workload and Actuator Scheduling

Abstract

[0001] A system optimization engine treats thermal, electrical, mechanical, and software telemetry as waveform data. Discrete logs (e.g., CPU utilization samples, pump duty cycles, fan PWM, inlet/outlet temperatures, and network request intervals) are transformed into continuous time-domain signals and then analyzed in the frequency domain. A spectral monitor identifies destructive interference patterns in which asynchronous periodic processes amplify heat, cause power-supply ripple, or induce latency spikes. A phase-shift controller applies bounded micro-delays to selected processes (e.g., workload scheduling offsets, pump cycle offsets, fan ramp timing, and DVFS/clock-gating windows) to align operational peaks into a constructive interference state that reduces peak thermal load and improves throughput, without changing the underlying task logic.

Technical Field

[0002] The disclosure relates to high-efficiency computation and thermal management, including optimization of cyber-physical systems, data-center and embedded thermal control, spectral analysis of operational telemetry, and control-theoretic scheduling and actuation that reduces energy consumption and latency by phase-aligning concurrent periodic processes.

Background

[0003] Modern computing and electromechanical platforms exhibit periodic behaviors across software and hardware layers, including scheduler quanta, request batching intervals, fan and pump duty cycles, power-conversion switching, and mechanical vibration modes.

[0004] When periodic processes are uncoordinated, their peaks can coincide in ways that increase instantaneous power draw, create thermal hotspots, amplify vibration, or induce queueing and latency spikes. These effects appear as emergent interference patterns that are difficult to predict from individual component logic.

[0005] Conventional optimizers focus on static parameter tuning (e.g., fixed fan curves) or workload placement (e.g., bin packing) but often ignore frequency-domain interactions among concurrent processes. There is a need for a runtime system that detects and mitigates destructive interference patterns while preserving the underlying functional logic of tasks and safety constraints.

Summary

[0006] An optimization engine continuously ingests telemetry streams from software and hardware components and converts discrete events and logs into continuous time-series signals suitable for spectral analysis.

[0007] A spectral monitor computes frequency-domain features (e.g., power spectral density, cross-spectral density, coherence, and phase) to identify destructive interference patterns among concurrent operations that correlate with heat, latency, or instability.

[0008] A phase-shift controller selects candidate processes and applies bounded micro-delays or timing offsets to align operational peaks into a constructive interference state that reduces peak heat and improves throughput, without changing the semantic ordering or correctness of tasks.

[0009] A dissonance alarm module predicts impending hardware failure or software gridlock by detecting persistent destructive patterns and trend shifts in spectral features, and emits maintenance or mitigation triggers.

Brief Description of the Drawings

[0010] - FIG. 1 illustrates an example architecture including telemetry collectors, a waveform transformation stage, a spectral monitor, a phase-shift controller, and controlled actuators.

[0011] - FIG. 2 illustrates an example of constructive interference achieved by phase-aligning two periodic processes.

Detailed Description

System Overview

[0012] In one embodiment, the system operates as a closed-loop optimizer that continuously observes a target system and applies small timing adjustments to reduce peak energy and improve performance. The optimizer may be implemented as a daemon, hypervisor service, container sidecar, embedded controller, or distributed control plane.

[0013] Telemetry sources may include software counters (CPU utilization, run-queue depth, garbage-collection pauses, network request intervals), hardware sensors (temperatures, voltages, currents), and actuator feedback (pump RPM, fan PWM, valve position). Telemetry is timestamped and normalized.

Digital Waveform Transformation

[0014] Discrete logs are converted into continuous-time signals by resampling to a uniform cadence, interpolating missing data, and applying windowing. For event streams, impulses may be convolved with a kernel to yield a continuous representation whose integral corresponds to event rate.

[0015] In one embodiment, the transformation produces both (i) a time-domain signal $x(t)$ and (ii) a short-time Fourier transform $X(f, t)$ to capture non-stationary behavior. Alternative transforms include wavelet transforms for multi-resolution analysis.

Spectral Monitor and Interference Detection

[0016] The spectral monitor computes power spectral density (PSD) for each signal and cross-spectral density (CSD) between signal pairs. Coherence metrics are used to identify correlated periodicities across subsystems.

[0017] Destructive interference is detected when phase relationships and peak alignments correlate with negative outcomes such as thermal spikes, power ripple, or tail latency. The monitor may maintain an interference map linking pairs or groups of processes to dominant frequencies and phase offsets.

[0018] In one embodiment, the monitor computes a risk score R that combines: (i) coherence magnitude, (ii) peak concurrence probability, (iii) measured impact on KPIs (e.g., temperature delta, p99 latency), and (iv) persistence over time.

Phase-Shift Controller

[0019] The controller selects one or more candidate processes for micro-delay insertion subject to constraints. Constraints may include service-level objectives, deadlines, control-loop stability limits, and maximum permissible jitter.

[0020] Phase shifting may be applied by adjusting scheduler offsets, aligning batch timers, shifting pump and fan cycles, staggering compressor starts, or coordinating DVFS/clock-gating windows. Offsets may be on the order of microseconds to seconds depending on the controlled frequency band.

[0021] In one embodiment, the controller searches for an offset vector Δ such that the predicted peak concurrency is minimized while preserving task correctness, using gradient-free search, model predictive control, or reinforcement learning constrained by hard safety bounds.

Thermal and Energy Optimization Use Cases

[0022] Data center: align compute bursts with cooling capacity by phase-offsetting batch jobs and synchronizing pump cycles to smooth heat flux through heat exchangers.

[0023] Embedded systems: coordinate actuator vibrations and CPU bursts to avoid exciting structural modes, reducing noise and improving sensor fidelity.

[0024] Industrial HVAC: align compressor and fan cycles across zones to reduce simultaneous inrush current and power demand charges.

Dissonance Alarm and Predictive Maintenance

[0025] Persistent destructive interference at specific frequencies can indicate mechanical wear (e.g., bearing resonance), power-supply degradation (increased ripple), or software pathologies (lock contention, thundering herd).

[0026] The dissonance alarm monitors trends in spectral features and triggers alerts when patterns exceed thresholds, change abruptly, or correlate with known failure signatures. Alerts may be routed to maintenance systems and can include recommended mitigation actions such as re-phasing or derating.

Implementation Notes

[0027] Telemetry pipelines may use ring buffers with monotonic clocks to preserve ordering. Spectral analysis may be accelerated using FFT libraries and GPU offload.

[0028] To preserve safety, the optimizer should log all applied phase shifts, maintain rollback capability, and enforce invariants such as maximum offset magnitude and maximum rate of change of offsets.

[0029] The techniques may be implemented locally or in a distributed control plane that coordinates multiple nodes by exchanging phase and frequency summaries.

Example Claims (Non-Limiting)

[0030] 1. A computer-implemented method comprising: collecting telemetry from at least one software process and at least one physical actuator; transforming discrete telemetry into a continuous waveform representation; performing spectral analysis to identify a destructive interference pattern among concurrent operations; and applying a bounded phase shift to timing of at least one operation to reduce the destructive interference pattern while preserving task logic.

[0031] 2. The method of claim 1, wherein transforming comprises computing a short-time Fourier transform and deriving at least one coherence metric between two telemetry signals.

[0032] 3. The method of claim 1, wherein applying the bounded phase shift comprises micro-delaying a scheduler event, a batch timer, or an actuator duty cycle to align operational peaks.

[0033] 4. The method of claim 1, further comprising computing a dissonance risk score indicative of impending hardware failure or software gridlock and emitting a predictive maintenance alert.

[0034] 5. The method of claim 1, wherein the bounded phase shift is selected subject to constraints including at least one of a deadline constraint, a service-level objective, and a control-loop stability constraint.

Drawings

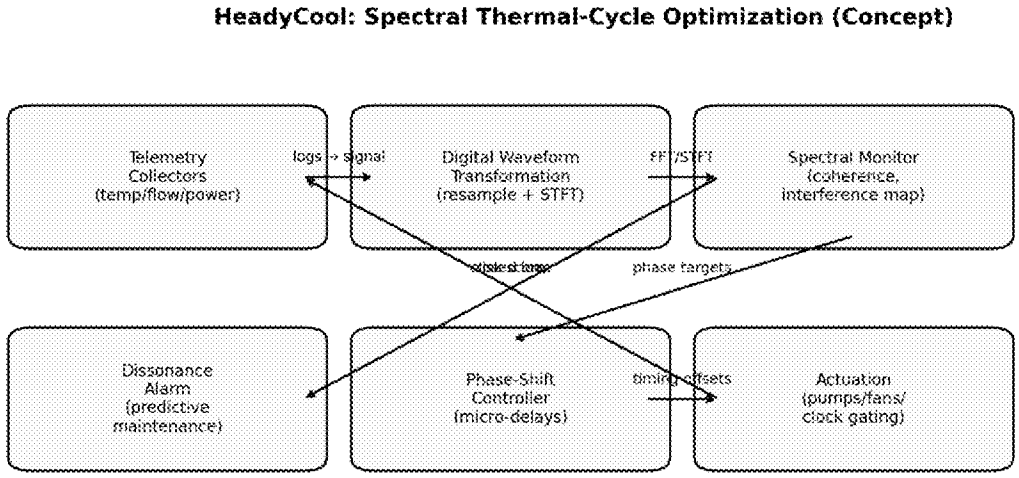

FIG. 1 — Spectral thermal-cycle optimization architecture.

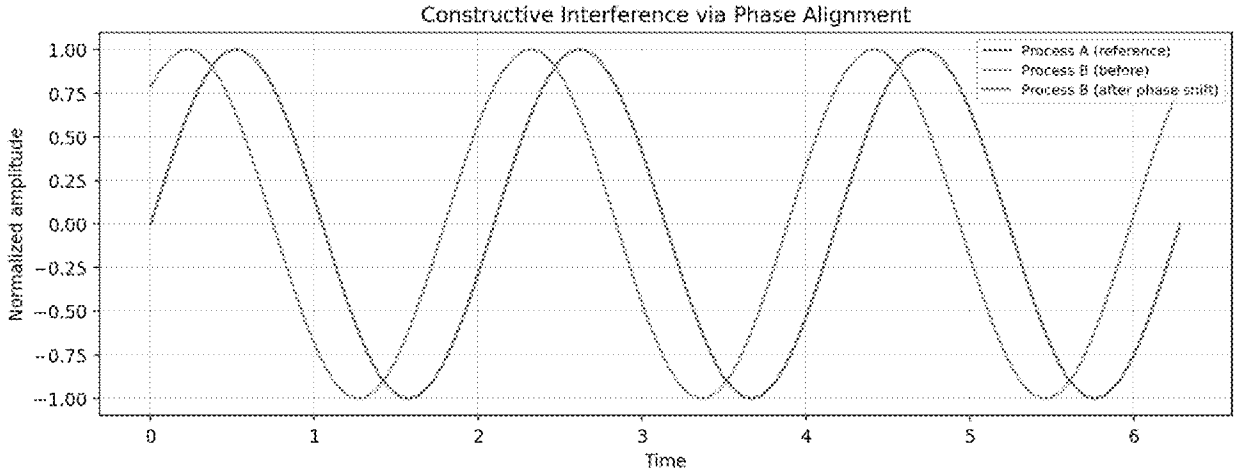

FIG. 2 — Constructive interference via phase alignment.

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PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/965,440	**01/22/2026 05:30:20 AM Z ET**	**-**

Title of Invention

Thermal System Efficiency Optimization via Spectral Analysis of Heat-Load Frequencies and Phase-Locked Cooling Actuation

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	1247	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74085730	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74085730.pdf	5	Application Data Sheet	111 KB
PPA-029_HeadyCool_SMALL_ENTITY_FINAL_v2-APP.TEXT.docx	7	Application body structured text document	346 KB

Warning: The specification paragraph numbering is not consecutive. Please review and revise if necessary.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)

generatedADS74085730.pdf	37A3CCBA1F7F9F449C47558692DEF55DBAC8F212F49785E93A4C236D753C0E181153761894D0C107D75C6BA95E354787D90A3A6A26CECB5685193D7F82FCD764
PPA-029_HeadyCool_SMALL_ENTITY_FINAL_v2-APP.TEXT.docx	E1023E44F4F23051B6E2EF648E7DDFF3CA145A49898F8CD2B60B02457C9E47C216FC3AFD99CBCA2427D7CA2996FA83DF1AE3A4550B44A85C11E5B8215598EBC5

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

DocCode - SCORE

Placeholder Sheet for Supplemental Content

Application Number: 63965440 Document Date: 01/22/2026

The presence of this form indicates that the following document type was received in electronic format on the date identified above. This document is stored separately from the typical case contents which are stored in image format.

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